

Vio Commerce x Vipps MobilePay ePayment — solution description

Partner: NewCo AS (Vio Commerce), Norway. **Product:** shoppable media — products sold inside articles, video pages, Vev pages and apps; the merchant's catalogue and orders stay in the merchant's shop (Shopify, WooCommerce) or in Vio. **Vio acts as a platform:** one integration, many merchants (sales units). Test sales unit used for this checklist and by the demo page: MSN 545865 (NewCo AS), through Vio's partner credentials.

Architecture

- `shopcart` (NestJS) owns the Vipps integration: payment creation, status, webhooks, capture / refund / cancel, receipts. `base-api` is the public edge (relays the webhook with its raw body and signature headers; exposes the dashboard routes). `graphql` exposes `InitPaymentVipps` and `GetVippsStatus` to the web SDK. `orders` creates the merchant order and tells `shopcart` when it ships or is cancelled.
- **Credentials per merchant:** *own* (the merchant's keys), *partner* (Vio's partner keys + the merchant's MSN, assigned when Vipps confirms the sales unit), or Vio's own sales unit.
- **Payment creation** (`POST /epayment/v1/payments`): `WALLET`, `WEB_REDIRECT`, `customerInteraction: CUSTOMER_NOT_PRESENT`, `paymentDescription` with the shop and order, `reference VIO-<checkout uuid>` (unique per sales unit, `^[a-zA-Z0-9-]{8,64}$`), `metadata` (`checkout`, `cart`, `seller`), `profile.scope` `name` `address` `email` `phoneNumber`, and for Express `shipping.fixedOptions` (the merchant's rates, brand/type mapped) or `shipping.dynamicOptions` (callback URL + token) with `allowedCountries`. In Express the amount is the goods only; Vipps adds the delivery option the shopper picks.
- **Completion — one path:** webhook, shopper return, status poll and a 10-minute sweep all call the same handler under a database lock; it reads the payment (`GET /payments/{reference}`), acts only on `AUTHORIZED`, creates the order once (idempotent), writes the shopper's details from `userDetails` / `shippingDetails`, sends the receipt (`POST /order-management/v2/ecom/receipts/{reference}`), and captures at once if the merchant chose capture-at-payment.
- **Webhooks:** all eight `epayments.payment.*.v1` events, signature verified (HMAC-SHA256 over `POST\n{path}\n{date};{host};{content-sha256}`, current and previous secret), deduplicated by event id; `authorized` completes, `aborted/expired/terminated/cancelled` close the checkout, `captured/refunded/cancelled` are recorded on the payment's history.
- **Polling:** the SDK polls `GetVippsStatus` on return; the sweep reconciles every open payment every 10 minutes (lost webhooks, closed tabs).
- **Capture:** at payment, at shipment (tracking number or order completion from the merchant's shop or the dashboard) or by hand from the order page; always $\leq$ what is reserved.
- **Refund / cancel:** order cancelled $\rightarrow$ release the reservation or refund the captured amount; partial refund from the order page; a paid reservation whose order could not be created after 24 h is released automatically and the failure logged.
- **Errors:** Vipps' error body is surfaced as `VippsApiError` (status, code, detail, trace id); the shopper sees a plain message ("Betalingen ble avbrutt eller feilet. Vennligst prøv igjen."), the merchant sees the reason in the dashboard; every failing call is logged on one line with endpoint, Vipps' error body (type, title, detail, traceId), the request headers (bearer token and subscription key masked) and the request body.

- **Headers** on every request: `Vipps-System-Name: Vio`, `Vipps-System-Version: 4.0.1` (the shopcart release), `Vipps-System-Plugin-Name: vio-commerce-shopcart`, `Vipps-System-Plugin-Version: 4.0.1`, plus an `Idempotency-Key` on create / capture / refund / cancel.
- **Branding**: Vipps MobilePay's own button component (`<vipps-mobilepay-button>`, from `cdn.vippsmobilepay.com`) on the product, the cart and the checkout.
- **Support**: the order page shows the payment as Vipps sees it (reserved / captured / refunded / released, reference, events) with the actions; support never needs the portal.

Log example (QA, 2026-10-08 17:04 UTC)

One line per failed call: endpoint, status, Vipps' title and detail, trace id, then the error body, the request headers (bearer token and subscription key masked) and the request body. This one is the receipt of the checklist's Create payment reference sent a second time (Vipps keeps the first):

```
[VippsConnector] POST /order-management/v2/ecom/receipts/VI0-ac20bc9a-d938-4982-b11e-e43a862e1434 → 409 Conflicting receipt – Receipt already exists (trace -) error {"title":"Conflicting receipt","status":409,"detail":"Receipt already exists","instance":"00-6c1b726bdd93992329e12f28c0db1577-7f48871c6101716e-01"} headers {"Accept":"application/json, text/plain, */*","Content-Type":"application/json","Authorization":"**","Ocp-Apim-Subscription-Key":"***","Merchant-Serial-Number":"545865","Vipps-System-Name":"Vio","Vipps-System-Version":"4.0.1","Vipps-System-Plugin-Name":"vio-commerce-shopcart","Vipps-System-Plugin-Version":"4.0.1","User-Agent":"axios/0.21.3","Content-Length":"355"} body {"orderLines":[{"name":"Lait Corporel Body Lotion","id":"5ULH0I5LAIYU0HR4","totalAmount":18900,"totalAmountExcludingTax":15120,"totalTaxAmount":3780,"taxPercentage":25,"unitInfo":{"unitPrice":18900,"quantity":"1","quantityUnit":"PCS"},"discount":0,"isReturn":false,"isShipping":false}],"bottomLine":{"currency":"NOK","tipAmount":0,"receiptNumber":"4511"}}
```

Demo

- Test page (Aller Media's Mote & Livsstil, Vio web SDK): <https://mote-livsstil-hub-vio.replit.app/skjonnhet/guider/vio-test-shoppable-favoritter> — open a product and tap the Vipps "buy now" button (Express), or pay with Vipps from the cart, with the Vipps test user.

NewCo AS - Vio Commerce, solution-description, 2026-10-08.